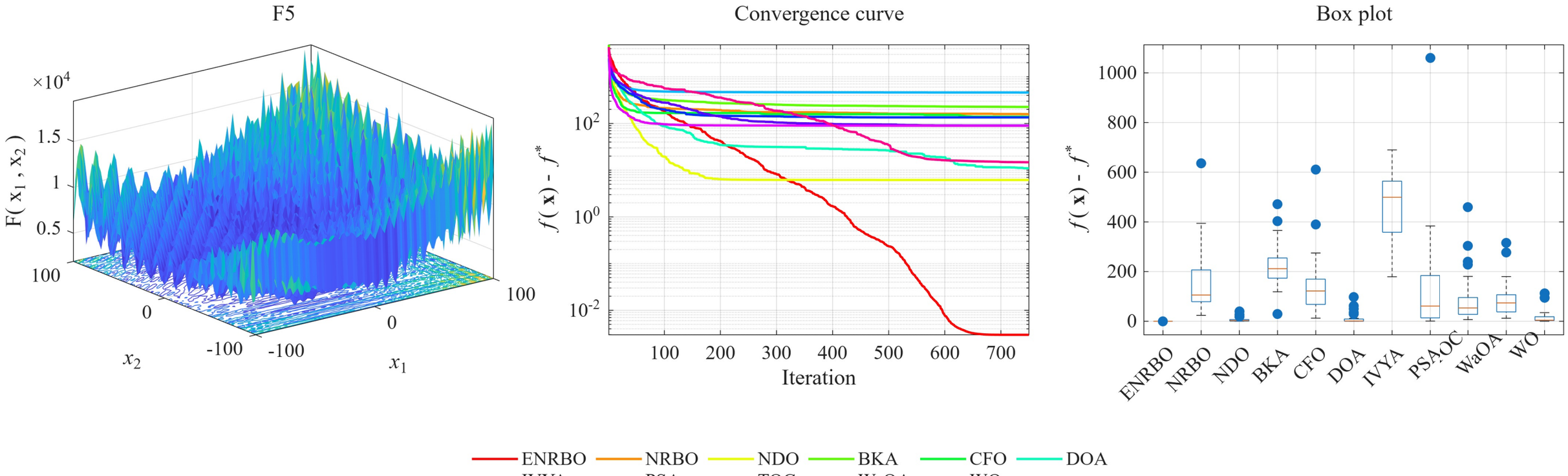

F5
×10^4
1.5
1
0.5
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
10^2
10^0
10^-2
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
1000
800
600
400
200
0
f(x) - f*
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSAOC
WaOA
WO
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
TOC
WaOA
WO